RELATIONAL ALGEBRA EXERCISES
SEM I – 2025/2026

QUESTION 1

Relational Schema:

BOOK (Bid, Title, Year, Category, Lid)
MEMBER (Mid, MName, City)
BORROW (Mid, Bid, DateBorrowed)
LIBRARIAN (Lid, LName, Branch)

BOOK

Bid	Title	Year	Category	Lid
1	AI Revolution	2021	Technology	1
2	History of Art	2019	Art	2
3	World Atlas	2020	Geography	1

MEMBER

Mid	MName	City
1	James	Johor Bahru
2	Aisha	Kuala Lumpur
3	Lim	Penang

LIBRARIAN

Lid	LName	Branch
1	Ravi	Main Branch
2	Sarah	Art Wing
3	Ismail	Southern Key

BORROW

Mid	Bid	DateBorrowed
1	1	2024-02-10
2	2	2024-03-15
3	3	2024-04-01

a) List all book titles and their categories.

b) List titles of books borrowed by members from 'Johor Bahru'.

c) Find librarians who do not manage any books.

d) List names of members who borrowed books published after 2020.

e) List librarian names and the titles of books they manage.

f) Product output for the given relational algebra:

i) $\pi_{Title,Year}$ (BOOK)

ii) $\pi_{Title,Category}$ ($\sigma_{Branch='MainBranch'}$(LIBRARIAN $\bowtie_{Lid=Lid}$ BOOK))

iii) $\pi_{MName,Title}$ (MEMBER$\bowtie_{Mid=Mid}$ (BORROW$\bowtie_{Bid=Bid}$ BOOK))

iv) πLName,Title,Branch (LIBRARIAN⋈Lid=Lid BOOK)

QUESTION 2

Relational Schema:

PRODUCT (Pid, PName, Category, Price, Sid)
SUPPLIER (Sid, SName, City)
CUSTOMER (Cid, CName, City)
PURCHASE (Cid, Pid, DatePurchase)

PRODUCT

Pid	PName	Category	Price	Sid
1	Rice	Grocery	30	2
2	Soap	Toiletry	10	1
3	Milk	Dairy	8	1

SUPPLIER

Sid	SName	City
1	Fresh Supply	Kuala Lumpur
2	Agro Mart	Johor Bahru

CUSTOMER

Cid	CName	City
1	Amir	Penang
2	Lina	Johor Bahru
3	Tan	Kuala Lumpur

a) List all product names and their prices.

b) List names of suppliers located in 'Kuala Lumpur'.

c) Find suppliers who do not supply any products.

d) List product names bought by customers from 'Penang'.

e) List product names and supplier names together.

QUESTION 3

Relational Schema:

FLIGHT (Fid, FName, Origin, Destination, Aid)
AIRPORT (Aid, AName, City)
PASSENGER (Pid, PName, Nationality)
BOOKING (Pid, Fid, SeatNo)

FLIGHT

Fid	FName	Origin	Destination	Aid
1	AK101	Kuala Lumpur	Tokyo	1
2	MH204	Penang	Seoul	2
3	SQ555	Singapore	London	3

AIRPORT

Aid	AName	City
1	KLIA	Kuala Lumpur
2	Penang Airport	Penang
3	Changi	Singapore

PASSENGER

Pid	PName	Nationality
1	Hiro	Japan
2	Siti	Malaysia
3	John	UK

a) List all flight names and their destinations.

b) List flight names that depart from 'Kuala Lumpur'.

c) Find airports that have no flights departing.

d) List names of passengers who booked flights to 'Tokyo'.

e) List flight names and airport names they belong to.

Product output for the given relational algebra:

i) $\pi_{FName,Destination}$ (FLIGHT)

ii) $\pi_{FName,City}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

iii) $\pi_{FName,Origin,Destination,Aname}$ (FLIGHT$\bowtie_{Aid=Aid}$ AIRPORT)

iv) $\pi_{AName,City}$ ($\sigma_{City='Penang'}$ (AIRPORT))

QUESTION 4

Relational Schema:

COURSE (Cid, CName, Category, Year, Tid)
TEACHER (Tid, TName, Expertise)
STUDENT (Sid, SName, Country)
ENROLL (Sid, Cid, Grade)

COURSE

Cid	CName	Category	Year	Tid
1	Python Basics	Programming	2023	1
2	Data Mining	Data Science	2025	2
3	Web Design	IT	2024	1

TEACHER

Tid	TName	Expertise
1	Dr. Ali	Programming
2	Ms. Tan	Data Science
3	Prof. John	Cybersecurity

STUDENT

Sid	SName	Country
1	Aisha	Malaysia
2	Ken	Japan
3	Sara	Singapore

a) List all course names and their categories.

b) List courses taught by teachers whose expertise is 'Data Science'.

c) Find teachers who do not teach any course.

d) List names of students who enrolled in courses after 2021.

e) List student names, course names, and teacher names.

QUESTION 5

ARTIST

Arid	ArName	Country
1	Adele	UK
2	Jay Chou	Taiwan
3	Lana Del Rey	USA

ALBUM

Alid	AlName	Year	Arid
1	30	2021	1
2	Chemtrails	2021	3
3	Fantasy	2022	2

TRACK

Tid	TName	Duration	Alid
1	Easy On Me	3:44	1
2	White Dress	5:33	2
3	Mojito	3:05	3

USER

Uid	UName	City
1	Hana	Kuala Lumpur
2	Rizal	Johor Bahru
3	Ken	Tokyo

LISTEN

Uid	Tid	DatePlay
1	1	2024-06-01
2	3	2024-06-05
3	2	2024-06-10

a) List track names and their durations.

b) List track names from albums released after 2020.

c) Find artists who do not have any albums.

d) List user names who listened to tracks by artist 'Adele'.

e) List artist names, album names, and track names.

QUESTION 6

Relational Schema:

EMPLOYEE (Eid, EName, City, Did)
DEPARTMENT (Did, DName, Location)
PROJECT (Pid, PName, Did)
WORKS_ON (Eid, Pid, Hours)

a) List employee names and their department IDs.

b) List projects located in the same department as employee 'Amin'.

c) Find departments that do not have any projects.

d) List employees who work more than 10 hours on any project.

e) List employee names and project names they work on.

QUESTION 7

Relational Schema:

PATIENT (Pid, PName, Age, WardNo)
DOCTOR (Did, DName, Specialty, WardNo)
WARD (WardNo, WardName, Location)

a) List names of all doctors and their specialties.

b) List names of doctors working in the 'Surgery' ward.

c) Find ward numbers that have no patients.

d) List names of patients treated by doctors in the 'Cardiology' ward.

e) List names of doctors and patients sharing the same ward.

QUESTION 8

Relational Schema:

MOVIE (Mid, Title, Year, Genre, Sid)
ACTOR (Aid, AName, Gender, Mid)
STUDIO (Sid, SName, City)

a) List movie titles and release years.

b) List titles of movies produced by 'Studio 7'.

c) Find studios that have not produced any movie.

d) List male actors who acted in movies after 2015.

e) List actor names, movie titles, and genres.

QUESTION 9

Relational Schema:

EMPLOYEE (Eid, EName, City, Did)
DEPARTMENT (Did, DName, Location)
PROJECT (Pid, PName, Did)
WORKS_ON (Eid, Pid, Hours)

a) List employee names and their department IDs.

b) List projects located in the same department as employee 'Amin'.

c) Find departments that do not have any projects.

d) List employees who work more than 10 hours on any project.

e) List employee names and project names they work on.